

HAT-P-12b

R-1032

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-12_160211 · created 2026-07-30T06:23:43+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2457429.8294 +/- 0.0017 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.104 +/- 0.019
Transit depth (Rp/Rs)^2	1.07 +/- 0.38 [%]
Orbital Inclination (inc)	90.0 +/- 1.12
Airmass coefficient 1 (a1)	0.971 +/- 0.016
Airmass coefficient 2 (a2)	0.0217 +/- 0.0027
Scatter in the residuals of the lightcurve fit is	1.7 %
Best Comparison Star	#2 - [67, 164]
Optimal Method	PSF photometry
Transit Duration (day)	0.0939 +/- 0.0037

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.104 ± 0.019 vs 0.1406 (deviation 1.79σ)
Duration fit vs published	0.0939 d vs 0.0975 d (deviation 0.9σ)
Mid-time O-C	-3.1 min
Depth SNR	5.1
Residual scatter	1.7 %

Light curve

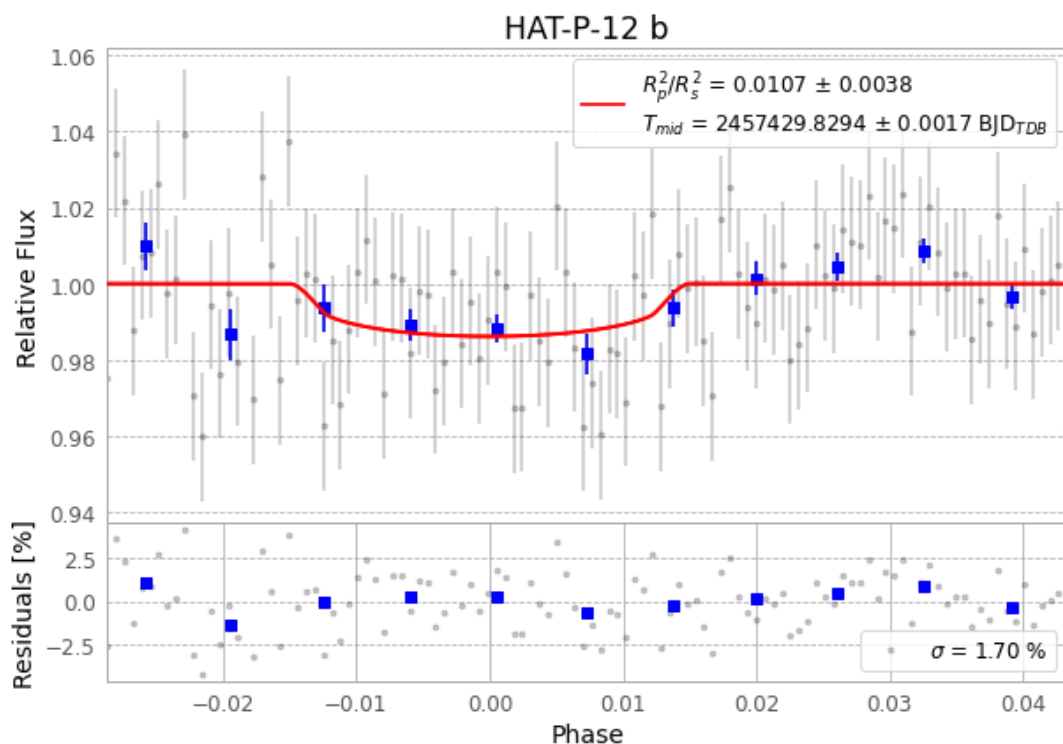


Figure 1 — FinalLightCurve_HAT-P-12 b_2016-02-10.png (SHA-256 5c8137e8f96f43d8...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [214, 123], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 086ca6c796fe495f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b65555c

Data provenance

112 FITS frames (74 MB), HATP-12160211053612.FITS → HATP-12160211110912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-12-160211.log (2f9999de2b2e...), FinalLightCurve_HAT-P-12 b_2016-02-10.png (5c8137e8f96f...), FinalLightCurve_HAT-P-12 b_2016-02-10.csv (ae143ed60fe4...)

Compute resources

Wall time 130.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 06:23Z.